SCHOOL OF COMPUTER SCIENCE & IT, DAVV

CS: 6418 – Advanced Database Management System

TEST-I **MTech I (CS/NM/SE)**

Duration: 1 Hour **Marks 20** **Date 28-09-26**

1. Why is DBMS preferred over a traditional file system? Consider a company has duplicate customer data in multiple files. What problems can occur, and how can DBMS solve them?
2. Design an ER diagram for a **College Management System** and Identify all entities, attributes, and relationships. Also convert ER diagram into relations.
3. Consider following relations
   STUDENT (SID, SName, Age, Dept), ***COURSE*** (CID, CName, Credits, Dept),
   ENROLLMENT (SID, CID, Marks), ***FACULTY*** (FID, FName, Dept, Salary), ***TEACHES*** (FID, CID)

   Write relational algebra for following:
   - Find the names of faculty members working in the IT department.
   - Find the names of students whose marks in any course are greater than 80.
   - Find the student IDs of students who belong to the CSE department but are not enrolled in any course.
   - Find the departments that have both students and faculty members.
   - Find the names of students who are enrolled in all courses offered by the CSE department.

   Write relational calculus for following:
   - Find the names of all students belonging to the CSE department.
   - Find the names and ages of students whose age is greater than 21.
   - Find the names of courses having credits greater than 3.
   - Find the student names and marks of students who scored more than 80.
   - Find the names of students who are enrolled in both C101 and C102.